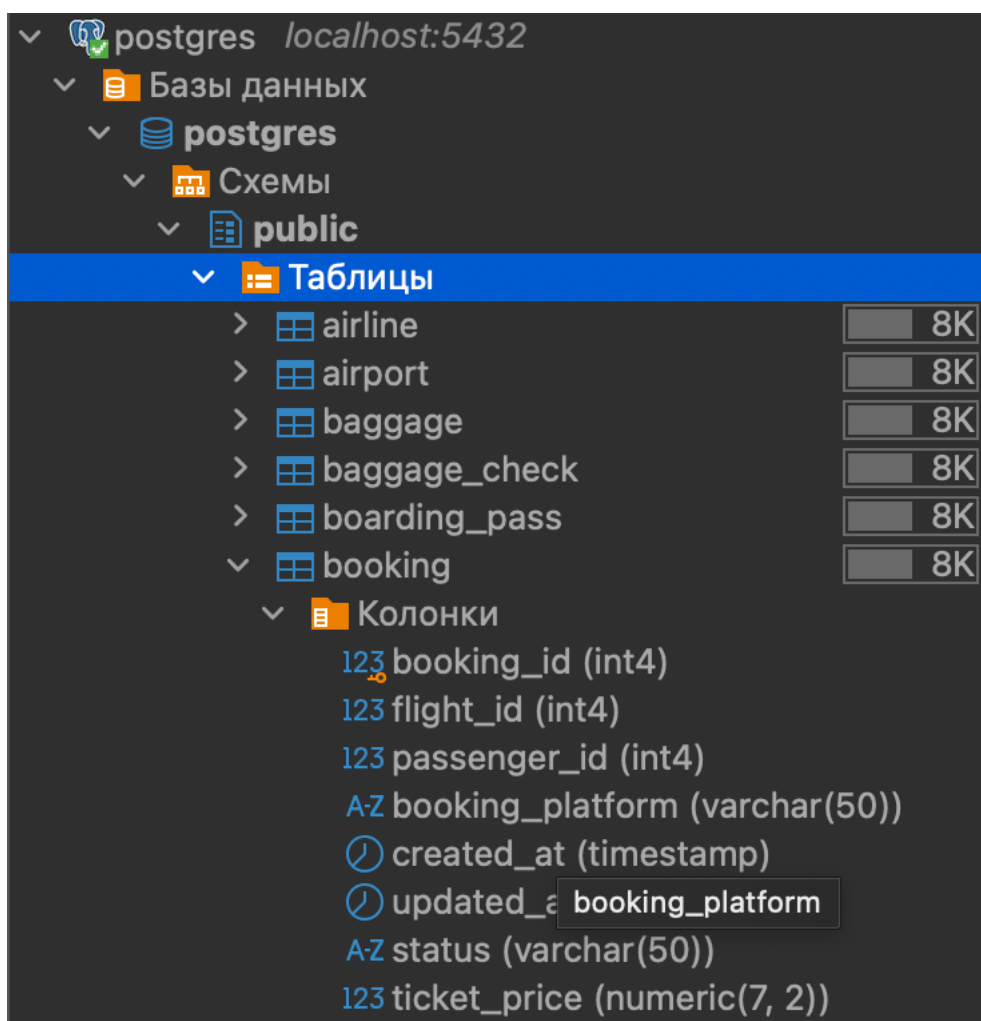


Step 1. 1NF Problem: Non-atomic values in seat\_numbers (e.g., 12A, 12B). Solution: Split into individual rows. Result: Table with atomic columns and a composite primary key

Step 2. 2NF Functional Dependencies: booking\_id determines booking details; passport\_number determines passenger name. Solution: Separate passenger details into a distinct table to remove partial dependencies

Step 3. 3NF Functional Dependencies: flight\_number determines airline and airport names. Solution: Move non-key attributes into separate lookup tables (Airline, Airport) to remove transitive dependencies and use foreign keys in Flights and Booking

Step 4. Flights Table 3NF Check Conclusion: The Flights table is in 3NF because it is in 1NF and 2NF, all non-key attributes depend directly on flight\_id, and airline/airport details are stored in separate parent tables linked via foreign keys



| Таблицы |                                   |    |
|---------|-----------------------------------|----|
| >       | airline                           | 8K |
| >       | airport                           | 8K |
| >       | baggage                           | 8K |
| >       | baggage_check                     | 8K |
| >       | boarding_pass                     | 8K |
| ▼       | booking                           | 8K |
| >       | Колонки                           |    |
| >       | Ограничения                       |    |
| >       | Внешние ключи                     |    |
| >       | Индексы                           |    |
| ▼       | Зависимости                       |    |
|         | fk_booking_passenger (a)          |    |
|         | fk_booking_flight (a)             |    |
|         | fk_booking_flight_booking (n)     |    |
|         | fk_baggage_check_booking (n)      |    |
|         | fk_baggage_booking (n)            |    |
|         | fk_boarding_pass_booking (n)      |    |
|         | booking_booking_id_not_null (a)   |    |
|         | booking_flight_id_not_null (a)    |    |
|         | booking_passenger_id_not_null (a) |    |
|         | booking_booking_platform_not_null |    |
|         | booking_created_at_not_null (a)   |    |
|         | booking_updated_at_not_null (a)   |    |
|         | booking_status_not_null (a)       |    |
|         | booking_ticket_price_not_null (a) |    |
|         | booking_pkey (a)                  |    |
|         | booking (i)                       |    |

Соединения | Проекты

Фильтровать соединения по имени

- postgres localhost:5432
  - Базы данных
    - postgres
      - Схемы
        - public
          - Таблицы
            - airline 8K
            - airport 8K
            - baggage 8K
            - baggage\_check 8K
            - boarding\_pass 8K
            - booking
              - Колонки
              - Ограничения
              - Внешние ключи
              - Индексы
              - Зависимости
                - fk\_booking\_passenger (a)
                - fk\_booking\_flight (a)
                - fk\_booking\_flight\_booking (n)
                - fk\_baggage\_check\_booking (n)
                - fk\_baggage\_booking (n)
                - fk\_boarding\_pass\_booking (n)
                - booking\_booking\_id\_not\_null (a)
                - booking\_flight\_id\_not\_null (a)
                - booking\_passenger\_id\_not\_null (a)
                - booking\_booking\_platform\_not\_null
                - booking\_created\_at\_not\_null (a)
                - booking\_updated\_at\_not\_null (a)
                - booking\_status\_not\_null (a)
                - booking\_ticket\_price\_not\_null (a)
                - booking\_pkey (a)
                - booking (l)
            - Ссылки
            - Секции таблиц
            - Триггеры
            - Правила
            - Policies
          - booking\_flight 8K
          - flights 16K
          - passengers 8K
          - security\_check 8K
        - Внешние таблицы
        - Представления
        - Мат. представления

select \* from passengers;

passengers 1

Введите SQL выражение чтобы отфильтровать результаты

|   | passenger_id | first_name | last_name | date_of_birth | gender |
|---|--------------|------------|-----------|---------------|--------|
| 1 | 1            | Ali        | Zhan      | 2000-05-15    | Male   |
| 2 | 2            | Aruzhan    | Kassymova | 1995-10-20    | Female |

Значение X

Обновить | Save | Cancel | Экспорт данных ... | 200 | 2

GMT+05:00 | Запись | Инт. вставка | 1 : 26 : 25 | Sel: 147 | 3

postgres localhost:5432

- Базы данных
  - postgres
    - Схемы
      - public
        - Таблицы
          - airline 8K
          - airport 8K
          - baggage 8K
          - baggage\_check 8K
          - boarding\_pass 8K
          - booking 8K
          - booking\_flight 8K
          - flights 16K
          - passengers 8K
          - security\_check 8K